

PRE-COVID (2019) **SOFIA** REAL ESTATE MARKET CONDITIONS ANALYSIS

- Riste Petrov

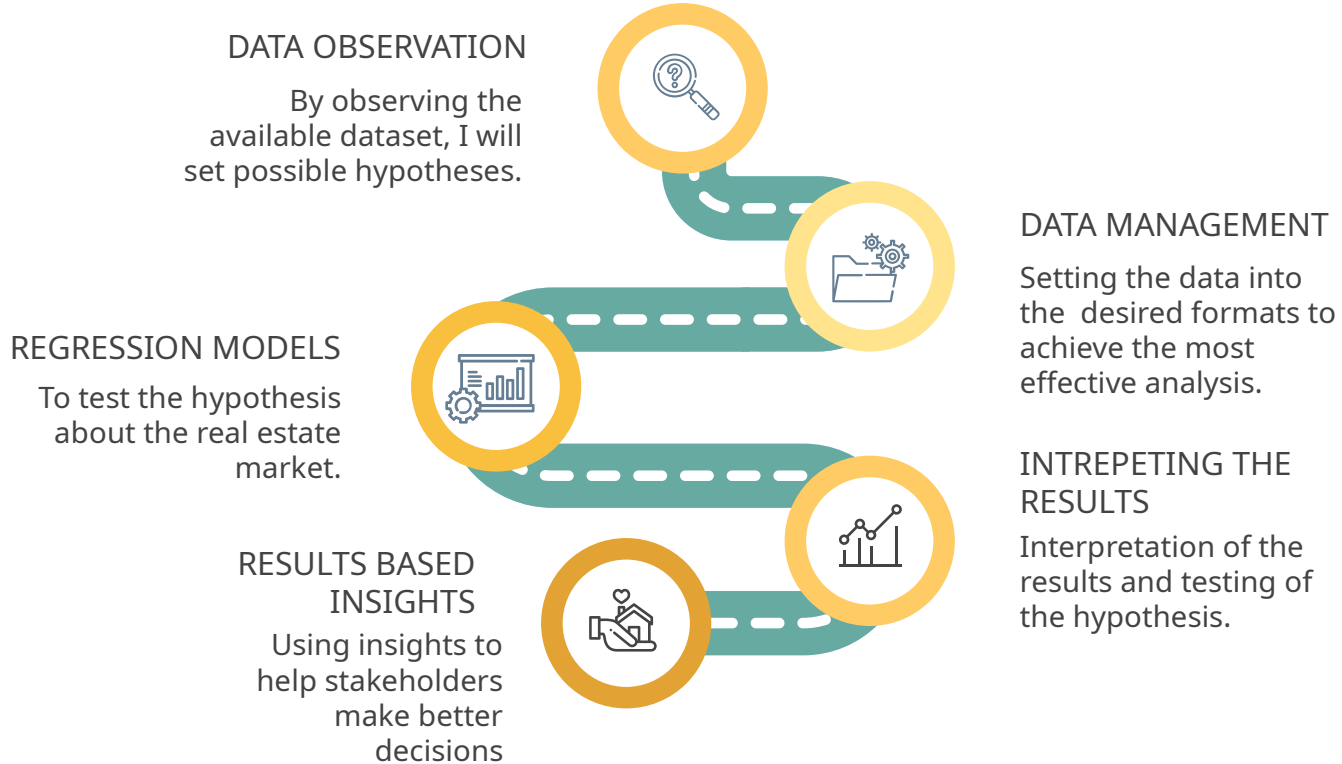


Why do research in real estate market conditions?

- Investment concentration, most people's most important financial investment decision of their life.
- Potential real estate bubble, not just in Bulgaria or the region but worldwide phenomena.
- Previous database (dated at 2004)'s fault of using different methodology of locations and their influence.



STAGES OF THE ANALYSIS



Variables Legend: (Name – operation – description) * - modified, X – excluded, T – transformation, D – dummy, N – none

Type – *, D - type of property

Regiid – X – region of the property

Townid – X – town of the property

Quartid - *, T – the neighborhood of the property

Price (Euro) – T (ln) – the advertised price for the property

Euro/sq.m. – t (ln) – the price per sq. meter

Area – T (mean,med per neig.) – area/size of the property

Constr. - *, D - type of construction of the property

Year – t (mean med per neig.) – construction year

Floor – N – floor of the property

TEC - * - central heating variable (ДА/HE) – transformed to 1/0

Published or Renewed & Valid date – * – a made a operational variable from both of them named Listing days.

Seller – D – Either Agency or Individual (personal_seller)

P.S If there was data if the apartment had furniture and was renovation would have helped with the models specification.



Data Analysis

Feature Engineering

The following data management was made:

Studio, Big Property & Story of a House were introduced to the type variable.

Various transformation made to Quartid explained in the next slide.

Brick, EPK, Pannel dummies were introudced.

Old dummy was set if the year of construction was after 2004.

Listing time was created but had limited usage in models, which could be based on the limitations of the messurment method, not the inherent meaning of it.



Quartid dummies - calculation

A prop map of Sofia's neighborhoods was done in Excel.

Each square is 500m x 500m

Using the map, each neighborhood's central point was calculated and set to be the NS = [-9,9] and WE = [-9,9] coordinate.

Distance from Center = $\sqrt{NS^2 + WE^2}$

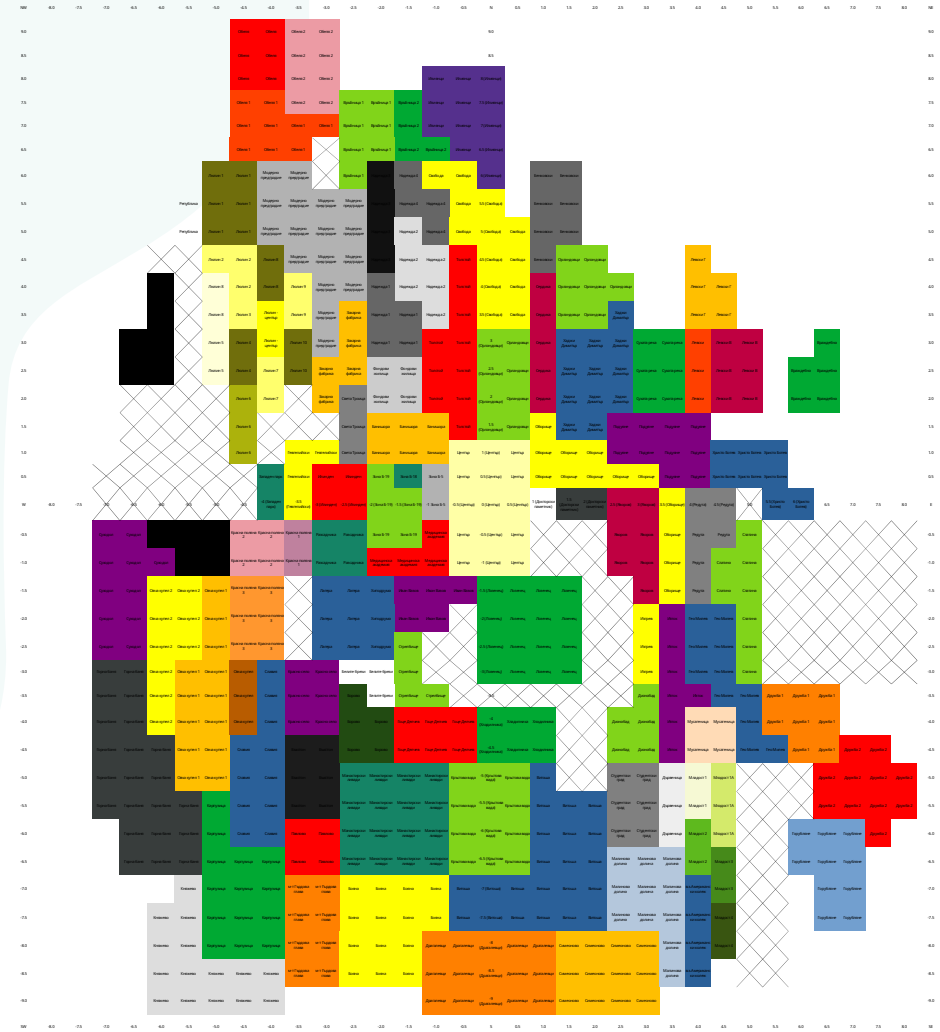
When compiling the map each neighborhood was assigned if it had metro access or not.

NS formula for Geo Milev

$$=-(\text{ROUND}((\text{ROW}(\$MAP.AA25) + \text{ROW}(\$MAP.AD30)) / 2, 1) - 21) / 2$$

WE formula for Geo Milev

$$=(\text{ROUND}((\text{COLUMN}(\$MAP.AA25) + \text{COLUMN}(\$MAP.AD30)) / 2, 1) - 19) / 2$$



Quartid Feature Engineering

LUX – Luxurious Location

PARK – attractive parks within walkable distance (500 – 1000 meters)

- Boris Garden, South, Western Park

MALL – shopping mall within walkable distance.

- Park Center, Bulgaria Mall

- Western Mall, Mall of Sofia

- Serdika Mall, The Mall, Ring Mall

- **KIDS** – number of kids in all kindergartens in the neighborhood

- **GARTEN (mean & median near 2)** – number of kindergartens in the neighborhood

- **KG_RATIO** – density of children per number of kindergartens in the neighborhood

- **KG (70% is 1)** – logical value if there is a kindergarten in the neighborhood. Kindergarten data:



Regression Analysis

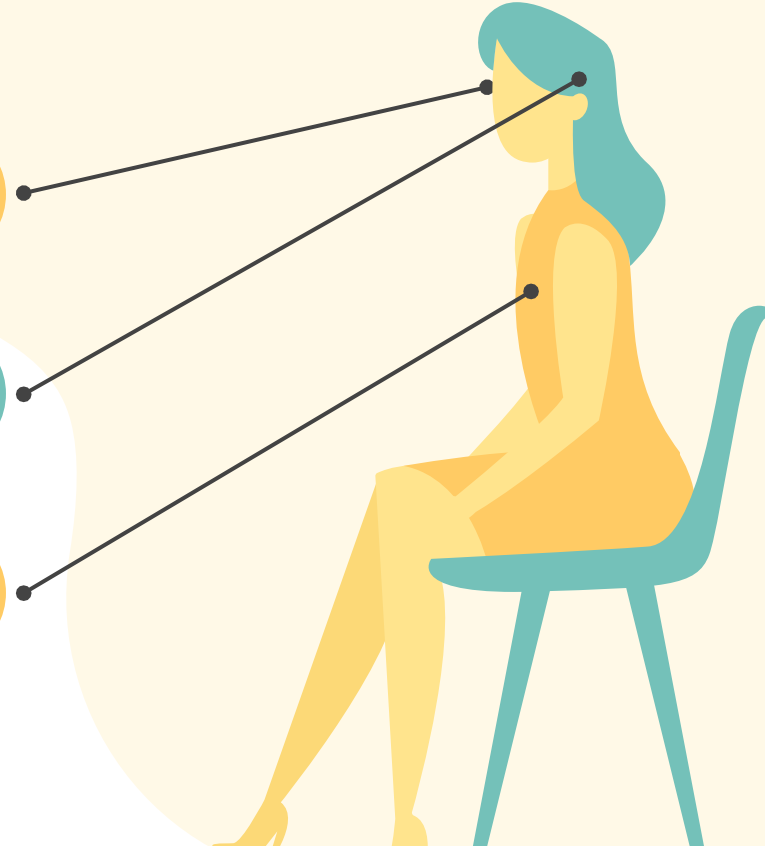
Over / Under priced logit
Using Over/Under priced
variables to
stimulate a real estate
professional.



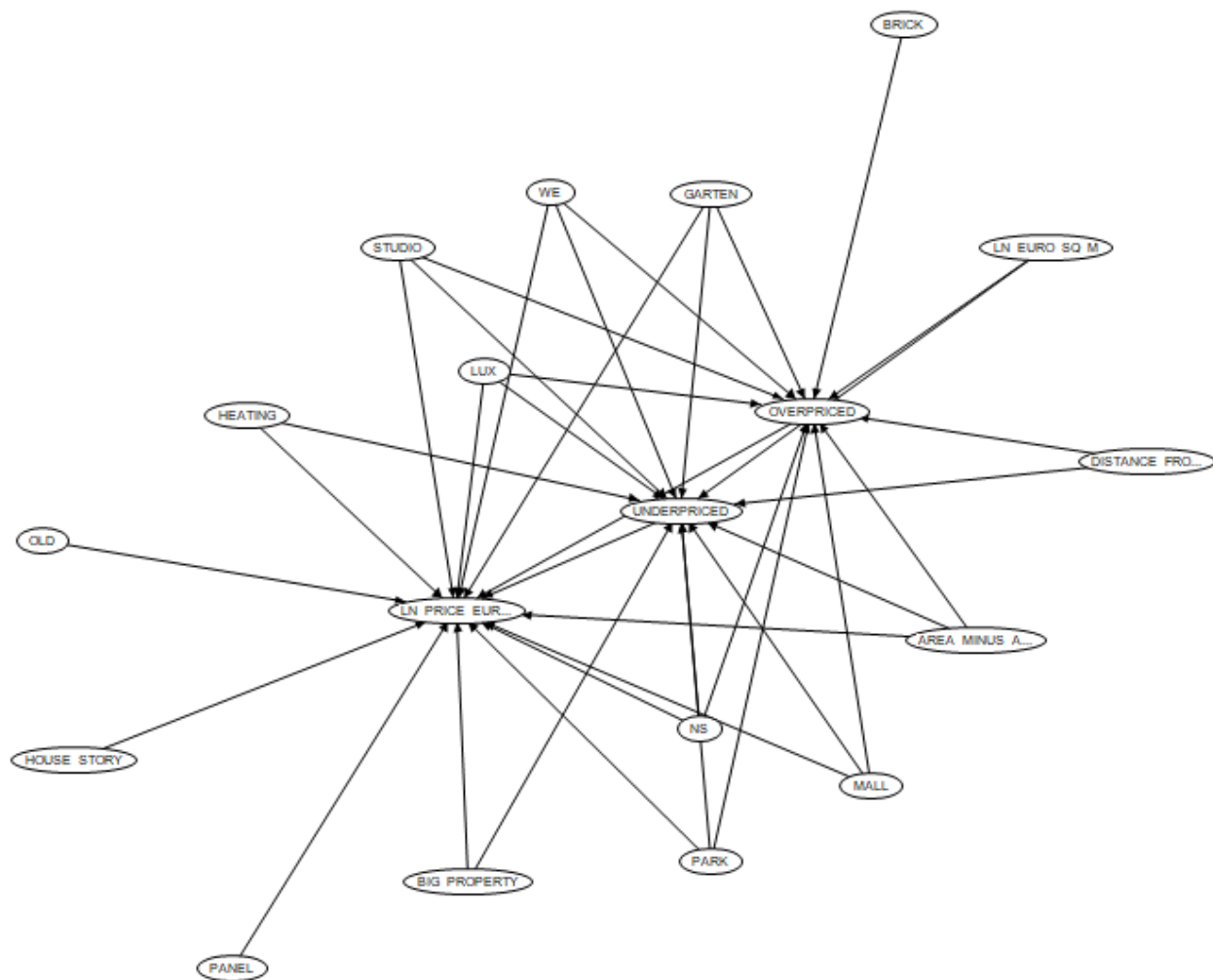
OLS Regression
Predicting $\ln(\text{price})$ using
exogenous regressors.



Structural model
Predicting $\ln(\text{price})$ using
layman-accessible regressors.



Structural Model



Under & Overpriced Definitions & Logit

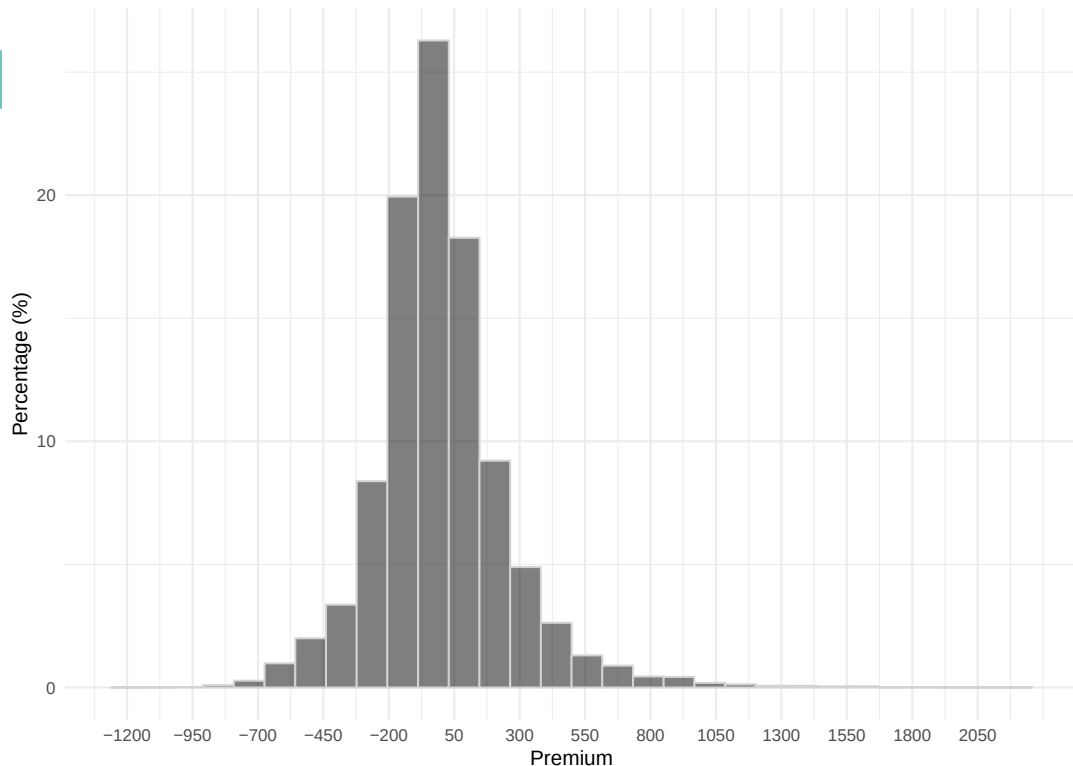
When predicting them as real estate professional (without any moral hazard) would have done it was considered **maximization** of Specifiacion (99%) or **minimization** of FPR (1%)

OVERPRICED – half standard deviation (**127.10 EUR**) over the median price of the square per meter in the neighborhood in which the property is located in.
(around 22% of properties)

FAIRLY PRICED – around 50% of the properties

UNDERPRICED – half standard deviation (**127.10 EUR**) under the median price of the square per meter in the neighborhood in which the property is located in.
(around 26% of properties)

Distribution of Premium



	MCFADDEN R2	AUC	Better than coinflip	MAX ACCURACY	ACCURACY AT FPR 1%	TRADEOFF at FPR 1%	CUT OFF probability for FPR 1%
OVERPRICED	0.6142	0.4457	5.43%	91.15%	88.92%	-2.24%	0.80
UNDERPRICED	0.5367	0.5747	7.47%	89.79%	88.43%	-1.36%	0.71

OLS regression versus Structure Model

Table 7: Model Coefficients and Statistics

Variable	Coefficient	<u>Model Coeff</u>	Std. Error	<u>Model STD</u>	t-Statistic	<u>Model t-stat</u>	Prob.	<u>Model Prob.</u>
C	11.2650	<u>11.2757</u>	0.0018	<u>0.0016</u>	6221.4476	<u>7063.2024</u>	0	<u>0</u>
AREA_MINUS_AVG_Q	0.0085	<u>0.0087</u>	0.0000	<u>0.0000</u>	237.7959	<u>273.2812</u>	0	<u>0</u>
NS	-0.0273	<u>-0.0271</u>	0.0002	<u>0.0002</u>	-131.8707	<u>-132.4238</u>	0	<u>0</u>
WE	0.0035	<u>0.0033</u>	0.0003	<u>0.0003</u>	10.5901	<u>10.8713</u>	3.48E-26	<u>1.67E-27</u>
GARTEN	0.0282	<u>0.0313</u>	0.0005	<u>0.0005</u>	55.0116	<u>68.1349</u>	0	<u>0</u>
HEATING	0.0392	<u>0.0235</u>	0.0018	<u>0.0017</u>	21.3752	<u>14.0032</u>	5.15E-101	<u>1.73E-44</u>
PANEL	-0.0369	<u>-0.0274</u>	0.0039	<u>0.0036</u>	-9.4984	<u>-7.6764</u>	2.20E-21	<u>1.66E-14</u>
BIG_PROPERTY (EXCLUDED)	-0.0185	NA	0.0053	NA	-3.5108	NA	4.47E-04	NA
HOUSE_STORY	-0.0664	<u>-0.0627</u>	0.0123	<u>0.0113</u>	-5.3933	<u>-5.5575</u>	6.94E-08	<u>2.75E-08</u>
STUDIO	-0.3497	<u>-0.3298</u>	0.0036	<u>0.0035</u>	-97.0671	<u>-94.7580</u>	0	<u>0</u>
LUX	0.2694	<u>0.2726</u>	0.0022	<u>0.0019</u>	125.2937	<u>144.7262</u>	0	<u>0</u>
PARK	0.1372	<u>0.1396</u>	0.0018	<u>0.0017</u>	76.0076	<u>83.2489</u>	0	<u>0</u>
MALL	-0.0358	<u>-0.0370</u>	0.0020	<u>0.0018</u>	-17.7563	<u>-20.2480</u>	2.28E-70	<u>7.15E-91</u>
OVERPRICED (ENDOGEN)	0.2538	<u>0.3493</u>	0.0020	<u>0.0021</u>	125.0280	<u>166.9589</u>	0	<u>0</u>
UNDERPRICED (ENDOGEN)	-0.2159	<u>-0.3256</u>	0.0019	<u>0.0020</u>	-113.3328	<u>-163.9697</u>	0	<u>0</u>
OLD (EXCLUDED)	0.0090	NA	0.0017	NA	5.4478	NA	5.12E-08	NA

Nominal Example – Model versus actual value

A baseline flat calculated by the model in the city center with an average area of 93.7 meters squared is priced around 130,119.81EUR. (without heating) or 131,636.55 EUR (with heating)

Which is underestimated within 5.5% and 4.5% of the median euro price per square meter for an average area flat of 93.7 meters.

Table 8: Model Comparison Metrics

Metric	Base	<u>Model</u>	Difference
R-squared	0.8016	0.8292	0.0276
Adjusted R-squared	0.8016	0.8292	0.0276
S.E. of regression	0.1956	0.1815	-0.0141
Sum squared resid	2,441.6283	2,102.0733	-339.5551
Log likelihood	13,579.9087	18,358.2952	4,778.3865
F-statistic	17,191.5111	23,834.1718	6,642.6607
Prob(F-statistic)	0.0000	0.0000	0.0000
Prob(Wald F-statistic)	0.0000	0.0000	0.0000
Mean dependent var	11.4878	11.4878	0.0000
S.D. dependent var	0.4392	0.4392	0.0000
Akaike info criterion	-0.4251	-0.5749	-0.1498
Schwarz criterion	-0.4228	-0.5729	-0.1501
Hannan-Quinn criterion	-0.4244	-0.5742	-0.1499
Durbin-Watson stat	1.6314	1.6243	-0.0070
Wald F-statistic	15,293.1629	22,274.9111	6,981.7481



Limitations

Measurement – all the measurements are based on the neighborhood which are inaccurate and assume that the used sample has a neutralization effect of intra neighborhood distribution of the observations

Multicollinearity – Potential problem with the prediction whether the property is over/under property using the natural log of price per square meter. (Not crucial problem as it has different distribution of the premium, see appendix)

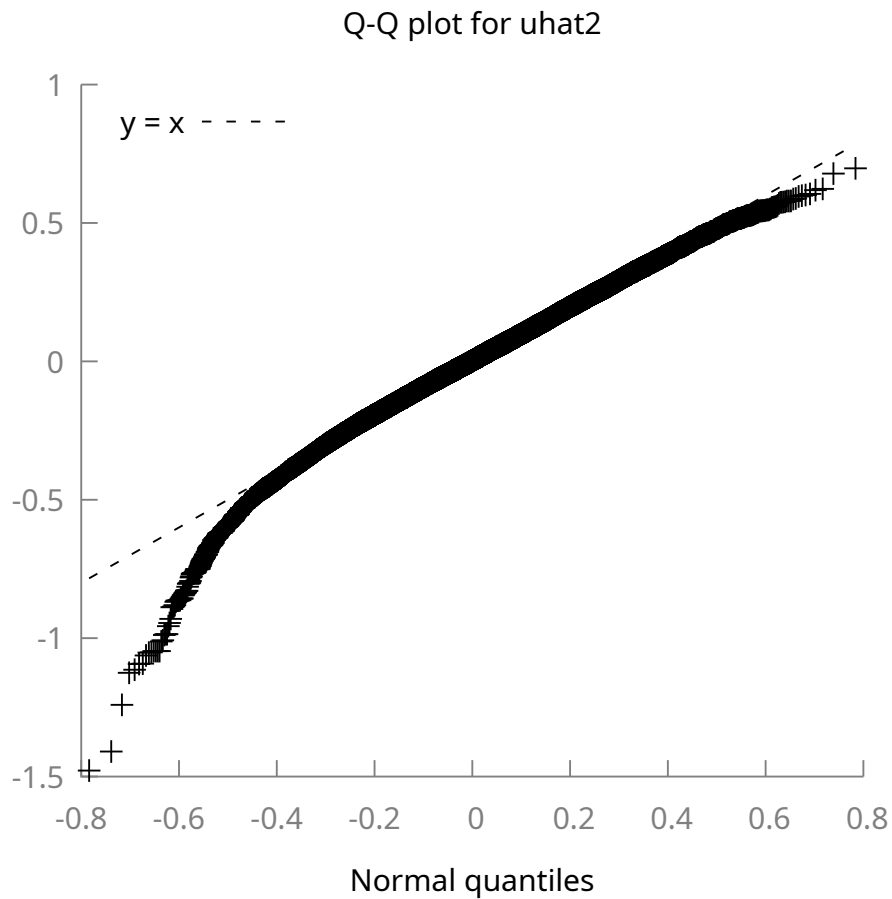
Heteroscedasticity – is present and HC standard errors were used as a remedy.

Misspecification – all regression fail the Ramsey Reset test implying that the models are theoretically misspecified.

Adynamic model – the model is based on the snapshot of 2019, no panel data analysis to see the effects through time and potentially in the future.

Residual QQ Plot

Normality of residuals is **not** present.



Quick Summary



Location

Location of the properties is influenced approximately by -3% for every 1km - North off the Center

Luxurious Location adds about 27%

A shopping center in the neighborhood reduces the price by around -3%



Property Classification

Panel type properties have a discount of around -3%

House stories have a discount of around -6%

Studios tend to have a discount of -32%



Central Heating, Park & Kindergarten

Central Heating increases the price by around 2%

Attractive park increases the price by around 13%

Each kindergarten in the neighborhood increases the price by around 3%

Thank you for your attention

Q&A

- “Real estate is all about location, location, location.”

— **Baron** Harold Samuel , founder
of Land Securities (UK)



APENDIX

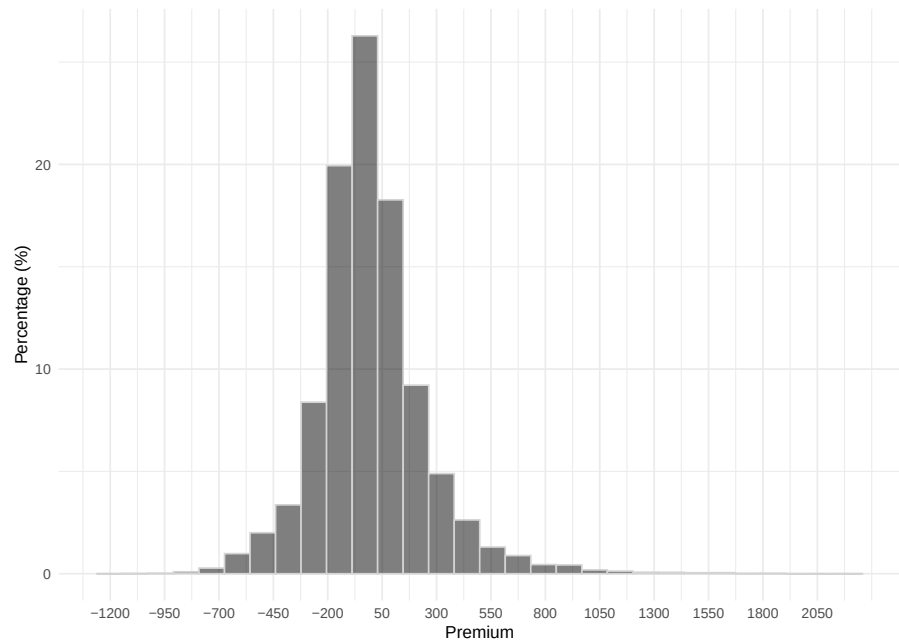


Table 1: Luxurious Location Neighborhoods by Average Euros per Square Meter

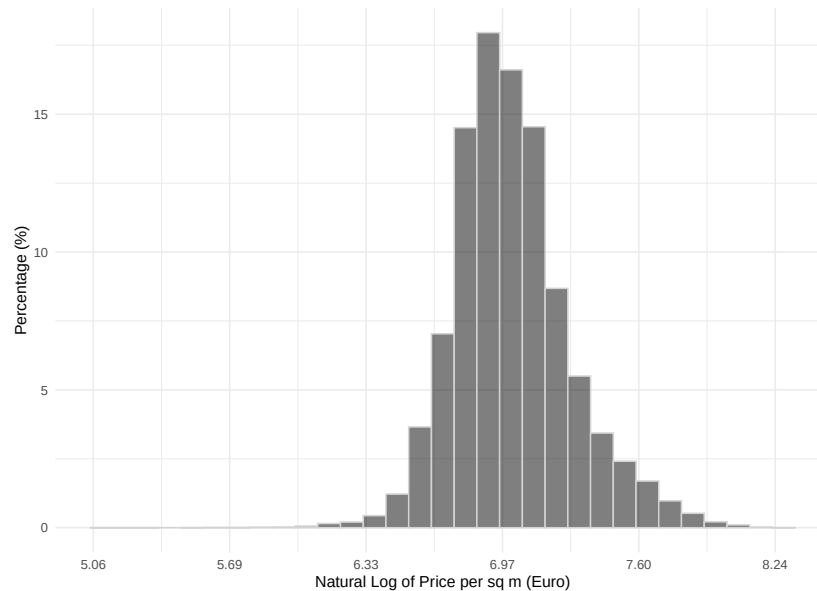
Neighborhood	Average Price (Euro/sqm)
Doktorski pametnik	2,232.14
Yavorov	1,914.66
Ivan Vazov	1,838.22
Izgrev	1,600.00
Iztok	1,595.48
Lozenec	1,582.21
Oborishte	1,510.66
Center	1,470.00
Boyana	1,155.17
Dragalevci	1,105.68
Manastirski livadi	1,073.17
Simeonovo	1,000.00

Premium & Price per Square Meter Minus Median of the neighborhood distributions

Distribution of Premium



Distribution of Natural Log of Property Prices per Square Meter in Sofia



Overpriced formula and regression output

$$\text{Probability of Overpriced} = \frac{1}{1 + e^{-(APREM+NS+WE+DFC+BRK+STUD+LUX+MLL+PRK+LNEUR)}}$$

Abbreviation explanation of products of coefficients and the respective values:

APREM - Area Premium (Observation Area - Average Area in Neighborhood)

NS - North-South coordinate for the neighborhood

WE - West - East coordinate for the neighborhood

DFC - Distance from Center

BRK - Brick dummy

STUD - Studio dummy

LUX - Luxurious location dummy

MLL - Shopping Mall dummy

PRK - Park dummy

GRT - Kindergarten number coeff

LNEUR - Natural Log Euro

Table 2: Regression Results for Overpriced

Variable	Coefficient	Std. Error	T-statistic	Probability
C	-146.4141	1.4149	-103.4802	0
AREA_MINUS_AVG_Q	-0.0020	0.0006	-3.3998	6.74E-04
NS	0.3035	0.0059	51.1905	0
WE	-0.2713	0.0071	-38.0121	0
DISTANCE_FROM_CENTER	0.7895	0.0129	61.2907	0
BRICK	-0.3284	0.0448	-7.3303	2.30E-13
STUDIO	0.8015	0.0634	12.6387	1.29E-36
LUX	-2.0820	0.0507	-41.0561	0
MALL	0.4036	0.0405	9.9591	2.30E-23
PARK	-0.5656	0.0467	-12.1223	8.04E-34
GARTEN	-0.3083	0.0113	-27.3200	2.45E-164
LN_EURO_SQ_M	20.1503	0.1953	103.1841	0

Underpriced formula and regression output

$$\text{Probability of Underpriced} = \frac{1}{1 + e^{-(APREM+NS+WE+DFC+STUD+LUX+MLL+PRK+HEAT+LNEUR)}}$$

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MLL - Shopping Mall dummy

PRK - Park dummy

GRT - Kindergarten number coeff

HEAT - Central Heating dummy

LNEUR - Natural Log Euro

Table 3: Regression Results for Underpriced

Variable	Coefficient	Std. Error	T-statistic	Probability
C	113.7220	1.0409	109.2509	0
AREA_MINUS_AVG_Q	0.0045	0.0005	8.8436	9.27E-19
NS	-0.4287	0.0069	-62.0574	0
WE	0.3047	0.0060	50.5009	0
DISTANCE_FROM_CENTER	-0.8541	0.0132	-64.5582	0
STUDIO	-0.2372	0.0659	-3.5992	3.19E-04
BIG_PROPERTY	0.3170	0.0502	6.3137	2.72E-10
LUX	2.7848	0.0443	62.8679	0
MALL	-0.7417	0.0345	-21.4875	2.04E-102
PARK	0.7540	0.0373	20.2247	5.93E-91
GARTEN	0.4103	0.0102	40.3932	0
HEATING	0.1878	0.0322	5.8373	5.30E-09
LN_EURO_SQ_M	-16.3306	0.1486	-109.9225	0

LOGIT - OLS Same Specification VIFs

Table 4: Overpriced OLS version: Coefficients and Variance Inflation Factors (VIF)

Variable	Coefficient	Variance	Uncentered VIF	Centered VIF
C	0.00		1,197.75	-
AREA_MINUS_AVG_Q	1.46E-09		1.20	1.20
NS	1.87E-07		2.70	1.96
WE	2.56E-07		1.49	1.47
DISTANCE_FROM_CENTER	7.70E-07		15.58	3.10
BRICK	1.30E-05		7.27	1.22
STUDIO	2.92E-05		1.20	1.12
LUX	1.26E-05		2.42	1.73
MALL	9.69E-06		2.63	1.57
PARK	1.16E-05		2.59	1.73
GARTEN	7.68E-07		3.47	1.65
LN_EURO_SQ_M	3.43E-05		1,131.19	1.67

Table 5: Underpriced OLS version: Coefficients and Variance Inflation Factors (VIF)

Variable	Coefficient	Variance	Uncentered VIF	Centered VIF
C	0.00		1,200.79	-
AREA_MINUS_AVG_Q	1.83E-09		1.30	1.30
NS	2.13E-07		2.64	1.93
WE	2.97E-07		1.49	1.48
DISTANCE_FROM_CENTER	8.78E-07		15.31	3.05
STUDIO	3.39E-05		1.20	1.12
BIG_PROPERTY	5.38E-05		1.15	1.11
LUX	1.46E-05		2.42	1.72
MALL	1.13E-05		2.63	1.57
PARK	1.35E-05		2.59	1.73
GARTEN	9.11E-07		3.55	1.69
HEATING	8.87E-06		2.20	1.25
LN_EURO_SQ_M	4.07E-05		1,155.90	1.71

Table 9: Coefficients Variance & VIF for the Non Structural Regression

Variable	Coefficient Variance	Uncentered VIF	Centered VIF
C	3.28E-06	5.9928	
AREA_MINUS_AVG_Q	1.28E-09	1.3038	1.2853
NS	4.28E-08	1.9879	1.4871
WE	1.07E-07	1.4282	1.3876
GARTEN	2.62E-07	3.4302	1.7793
HEATING	3.37E-06	2.4355	1.4097
PANEL	1.51E-05	1.1377	1.0894
BIG_PROPERTY	2.78E-05	1.0836	1.0507
HOUSE_STORY	1.51E-04	1.0219	1.0154
STUDIO	1.30E-05	1.2943	1.2243
LUX	4.62E-06	3.1867	1.9616
PARK	3.26E-06	2.0615	1.3237
MALL	4.07E-06	3.2687	1.8091
OVERPRICED	4.12E-06	1.4700	1.1638
UNDERPRICED	3.63E-06	1.5627	1.1797
OLD	2.72E-06	2.5712	1.1409

Table 10: Coefficients Variance & VIF for the Structural (Modeled) Regression

Variable	Coefficient Variance	Uncentered VIF	Centered VIF
C	2.55E-06	5.3497	
AREA_MINUS_AVG_Q	1.02E-09	1.2199	1.2055
NS	4.20E-08	2.0737	1.4287
WE	8.95E-08	1.3289	1.3133
GARTEN	2.11E-07	3.2106	1.6435
HEATING	2.82E-06	2.4059	1.3646
PANEL	1.27E-05	1.1076	1.0621
HOUSE_STORY	1.27E-04	1.0216	1.0152
STUDIO	1.21E-05	1.2500	1.1917
LUX	3.55E-06	2.9175	1.7481
PARK	2.81E-06	2.1704	1.3314
MALL	3.34E-06	3.0286	1.6684
OVERPRICED_CUTOFF	4.38E-06	1.3157	1.1185
UNDERPRICED_CUTOFF	3.94E-06	1.3571	1.1348